Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

PENDAHULUAN
TINJAUAN PUSTAKA
[backend=bibtex,sorting=none,style=numeric]biblatexref.bib
dashrule [named=NJUPurple]beamercolorthemestructure
[utf8]inputenc graphicx booktabs subfigure subfiles url amssymb
amsmath xcolor,colortbl [AutoFakeBold, AutoFakeSlant]xeCJK
[onlymath]beamerfontthemeserif
figs/cover.pdf
latexsym,amsmath,xcolor,multicol,booktabs,calligra
graphicx,pstricks,listings,stackengine wasysym

Your Title Your Title Your Title Your Title Your Title Your Title Your Title

figs/cover.pdf

Conference 2024

Your Institution

2025.2.19

About the Author()

figs/profile.jpg

LinaBell

Research Fellow in Nanjing University, a member of LAMDA.

figs/content.pdf

Latest works

- NeurPIS
- ICML
- ICRL
- TPAMI
- PRML

Outline

1. Preliminary
2. Related Work
3. Methodology

figs/content.pdf

4. Analyses
5. Experiments
6. Conclusion

Preliminary()

- **Learning Strategy**

  Optimization methods: Pointwise loss (binary cross-entropy, mean square error), pairwise loss (BPR, WARP), and softmax loss

figs/content.pdf

$$\mathcal{L}_0 = -\sum_{(u,i)\in O^+} \log \frac{\exp\left(\cos(\hat{\theta}_{ui})/\tau\right)}{\exp\left(\cos(\hat{\theta}_{ui})/\tau\right) + \sum_{j\in N_u} \exp\left(\cos(\hat{\theta}_{uj})/\tau\right)},$$

Outline

1. Preliminary
2. Related Work
3. Methodology

figs/content.pdf

4. Analyses
5. Experiments
6. Conclusion

Related Work()

SOTA debiasing strategies

- **Sample re-weighting methods** (e.g. IPS-CN)
  exploit the item popularity's inverse to re-weight loss of each instance.
- **Causal inference methods** (e.g. MACR, CausE)
  - specify the role of popularity bias in assumed causal graphs
  - mitigate the bias effect on the prediction.
- **Regularization-based frameworks** (e.g. Sam-reg)
  - Provides a tunable mechanism for controlling the trade-off between recommendation accuracy and coverage.
  - **Sam-reg** regularizes the biased correlation between user-item relevance and item popularity

figs/content.pdf

Outline

1. Preliminary
2. Related Work
3. Methodology
4. Analyses
5. Experiments
6. Conclusion

figs/content.pdf

BC loss

Methodology of BC Loss BC Loss()

- **BC Loss**

$$\mathcal{L}_{\mathrm{BC}} = -\sum_{(u,i)\in O^+} \log \frac{\exp\left(\cos(\hat{\theta}_{ui}+M_{ui})/\tau\right)}{\exp\left(\cos(\hat{\theta}_{ui}+M_{ui})/\tau\right)+\sum_{j\in N_u}\exp\left(\cos(\hat{\theta}_{uj})/\tau\right)}$$

M_{ui}: the bias-aware angular margin for the interaction (u, i)

figs/content.pdf

$$M_{ui} = \min\{\hat{\xi}_{ui}, \pi - \hat{\theta}_{ui}\}$$

- **Intuition**
  If a user-item pair is the hard interaction that can hardly be reconstructed by its popularity statistics, it holds a high value of ξ_{ui} and leads to a high value of M_{ui}. Henceforward, BC loss imposes the large angular margin M_{ui} between the negative item j and positive item i.

Outline

1. Preliminary
2. Related Work
3. Methodology

figs/content.pdf

4. Analyses
   - Geometric Interpretation
   - Theoretical Properties
5. Experiments
6. Conclusion

Analyses() Geometric Interpretation

- **Geometric Interpretation**
  User u with one observed item i and two unobserved items j and k.

figs/content.pdf



Analyses() Theoretical Properties

- **Theoretical Properties**

Bukti.

1. There exists an upper bound m, s.t.
$-1 < \cos(\hat{\theta}_{ui} + M_{ui}) \leq v_u^T v_i - m < 1$
figs 2.
3.
4.
5.
6.

□

Outline

1. Preliminary
2. Related Work
3. Methodology

figs/content.pdf

4. Analyses
5. Experiments
6. Conclusion

Experiments() Baselines & Datasets

Baselines

- Backbone: only use softmax loss
- IPS-CN: sample re-weighting methods
- CausE: bias removal by causal inference
- sam + reg: regularization-based framework
- MACR: bias removal by causal inference

figs/content.pdf

Datasets

	KuaiRec	Douban Movie	Tencent	Amazon-Book	Alibaba-iFashion	Yahoo!R3	Coat
#Users	7175	36,644	95,709	52,643	300,000	14382	290
#Items	10611	22,226	41,602	91,599	81,614	1000	295
#Interactions	1062969	5,397,926	2,937,228	2,984,108	1,607,813	129,748	2,776
Sparsity	0.01396	0.00663	0.00074	0.00062	0.00007	0.00902	0.03245

Outline

1. Preliminary
2. Related Work
3. Methodology

figs/content.pdf

4. Analyses
5. Experiments
6. Conclusion

Conclusion()

- **Contribution**
  - (Originality) Popular bias extractor has an intuitive geometric interpretation.
  - (Quality) Outperforms existing methods in various evaluation protocols.
  - (Clarity) Well-written and easy to understand. Theoretical proof is quite solid.

figs/content.pdf

- **Limitation**
  - The technical contribution of this paper is limited.It only proposes to employ an extra popularity-based predictor and combine the results with an existing CF modelhe2020momentum.
  - Overclaims the strength of the proposed BC loss in theoretical analysis. The geometric interpretability and the hard-negative mining ability are actually the same thing pmlr-v119-wang20k, yuan2021one

References I

[heading=none]


Acknowledgement

Thank you!

figs/content.pdf